

LECTURE 2

Counting the Infinite, and the Geometry of Distance: Countability and Metric Spaces

MATH S4061 | Introduction to Modern Analysis I

Thursday, 28 May 2026

No one shall expel us from the paradise that Cantor has created.

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- 2.1 Counting the infinite: countable and uncountable sets
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Overview. Lecture 1 built $\mathbb{R}$ as the complete ordered field and drew from completeness the Archimedean property (1.7.1) and the density of $\mathbb{Q}$ (1.7.2); the same property gives infima as readily as suprema (1.5.3). Two new questions open here. First, *how large* is $\mathbb{R}$? Cantor’s notion of countability lets us compare infinite sets, and his diagonal argument shows $\mathbb{R}$ is strictly larger than $\mathbb{Q}$ —uncountable, with $\mathbb{Q}$ a vanishingly “thin” subset. Second, what is the right language for *distance*? Metric spaces distil the Euclidean norm into three axioms, and the open sets they generate are the gateway to topology, where the rest of the course lives.

2.1 Counting the infinite: countable and uncountable sets

Definition 2.1.1

Injective, surjective, bijective

Let $f : X \rightarrow Y$ be a function.

- f is *injective* if $f(x) = f(x') \Rightarrow x = x'$ (distinct inputs give distinct outputs);
- f is *surjective* if every $y \in Y$ equals $f(x)$ for some $x \in X$;
- f is *bijective* if it is both. A bijection has a two-sided inverse $f^{-1} : Y \rightarrow X$.

Definition 2.1.2

Equinumerous sets; finite, countable, uncountable

Sets X and Y are *equinumerous*, written $X \sim Y$, if there is a bijection $f : X \rightarrow Y$. With $J_0 = \emptyset$ and $J_n = \{1, \dots, n\}$, a set X is

- *finite* if $X \sim J_n$ for some n , and *infinite* otherwise;
- *countable* if $X \sim \mathbb{N}$;
- *uncountable* if it is infinite and $X \not\sim \mathbb{N}$.

Proposition 2.1.3

Equinumerosity is an equivalence relation

The relation $\sim$ is reflexive, symmetric, and transitive.

Proof 2.1.3

Direct Proof that Equinumerosity is an Equivalence Relation.

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|---|--|
| <div style="border-left: 1px solid black; height: 100px; margin-left: -1px;"></div> | <p>(1) For reflexivity, the identity $\text{id}_X : X \rightarrow X$, $x \mapsto x$, is a bijection, so $X \sim X$.</p> <p>(2) For symmetry, if $X \sim Y$ via a bijection f, then $f^{-1} : Y \rightarrow X$ is a bijection, so $Y \sim X$.</p> <p>(3) For transitivity, if $X \sim Y$ via f and $Y \sim Z$ via g, then $g \circ f : X \rightarrow Z$ is a bijection (its inverse is $f^{-1} \circ g^{-1}$), so $X \sim Z$.</p> |
|---|--|

■

[Direct] bijections invert and compose, with $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Example 2.1.4

$\mathbb{Z}$ is countable

The identity shows $\mathbb{N}$ is countable. For $\mathbb{Z}$, interleave the signs:

$$f : \mathbb{N} \rightarrow \mathbb{Z}, \quad f(n) = \begin{cases} n/2 & n \text{ even,} \\ -(n+1)/2 & n \text{ odd,} \end{cases}$$

a bijection that lists $\mathbb{Z}$ as $0, -1, 1, -2, 2, -3, 3, \dots$

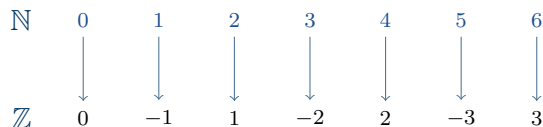


Figure 2.1.5. $\mathbb{N}$ COUNTS $\mathbb{Z}$. The bijection of 2.1.4 walks outward from 0 in alternating steps, reaching every integer exactly once.

Definition 2.1.6

Sequences and the listing criterion

A *sequence* in a set X is a function $f : \mathbb{N} \rightarrow X$, written $\{x_n\}_{n \in \mathbb{N}}$ with $x_n = f(n)$. Thus X is countable precisely when some sequence lists every element of X exactly once.

Example 2.1.7

$\mathbb{Q}$ is countable

Identifying a fraction with its pair of integers exhibits $\mathbb{Q}$ inside $\mathbb{Z} \times \mathbb{Z}$, which a diagonal sweep lists as a single sequence.

Proof 2.1.7

Proof by Diagonal Enumeration that $\mathbb{Q}$ is Countable.

- (1) Arrange the pairs (p, q) with $q \neq 0$ on a grid and traverse them along successive finite diagonals (each diagonal $|p| + |q| = \text{const}$ holds only finitely many pairs).
- (2) This runs through every pair; deleting pairs with a common factor and any repeats leaves a sequence that lists each rational exactly once.
- (3) Hence $\mathbb{Q} \sim \mathbb{N}$ (2.1.6).

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[Constr] cf. R 2.12–R 2.13.

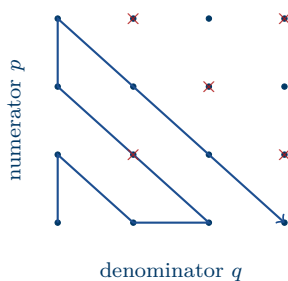


Figure 2.1.8. LISTING THE RATIONALS. Sweeping the lattice of pairs (p, q) along finite diagonals threads them into one sequence; pairs with a common factor (e.g. $\frac{2}{2}, \frac{2}{4}$) are skipped (2.1.7).

Proposition 2.1.9

An infinite subset of a countable set is countable

If X is countable and $A \subseteq X$ is infinite, then A is countable.

Proof 2.1.9

Proof by Recursive Selection of Indices that the Subset is Countable.

- (1) Write $X = \{x_n\}_{n \in \mathbb{N}}$, every element listed once (2.1.6).
- (2) Choose indices recursively: let n_0 be the least index with $x_{n_0} \in A$; having chosen $n_0 < \dots < n_k$, let n_{k+1} be the least index $> n_k$ with $x_{n_{k+1}} \in A$. Such an index exists because A , being infinite, cannot lie inside the finite set $\{x_0, \dots, x_{n_k}\}$, so some $x_m \in A$ has $m > n_k$.
- (3) The map $k \mapsto x_{n_k}$ is strictly increasing, hence injective, and reaches every element of A (each $a = x_m \in A$ is selected no later than stage m); so it is a bijection $\mathbb{N} \rightarrow A$, and $A \sim \mathbb{N}$.

■

[A10] the recursion is an induction, justified at each step by A being infinite.

Theorem 2.1.10

Cantor: the binary sequences are uncountable

The set $2^{\mathbb{N}} = \{f : \mathbb{N} \rightarrow \{0, 1\}\}$ of infinite binary sequences—equivalently, the subsets of $\mathbb{N}$ —is uncountable.

Proof 2.1.10

Proof by Cantor's Diagonal Argument that the Binary Sequences are Uncountable.

- (1) Suppose $2^{\mathbb{N}}$ were countable, listed as $a_1, a_2, a_3, \dots$, each $a_k = (a_k^1, a_k^2, a_k^3, \dots)$ a binary sequence.
- (2) Define $\bar{b} \in 2^{\mathbb{N}}$ by flipping the diagonal: $\bar{b}_n = 1 - a_n^n$.
- (3) For each n , $\bar{b}$ differs from a_n in its n th entry ($\bar{b}_n \neq a_n^n$), so $\bar{b} \neq a_n$ for every n .
- (4) Thus $\bar{b}$ appears nowhere in the list—contradicting that the list was all of $2^{\mathbb{N}}$. Hence $2^{\mathbb{N}}$ is uncountable.

■

[A8+A9] diagonalisation, by contradiction; cf. R 2.14.

a_1	0	1	1	0	1	0
a_2	1	1	1	1	0	1
a_3	0	0	0	1	0	1
a_4	0	1	1	0	1	0
a_5	1	0	0	0	0	1
$\bar{b}$	1	0	1	1	1	...

Figure 2.1.11. CANTOR'S DIAGONAL. Flipping the boxed diagonal entries yields a sequence $\bar{b}$ differing from each listed a_k in place k , so $\bar{b}$ is absent from the list (2.1.10).

Theorem 2.1.12

The interval $[0, 1]$ is uncountable

The interval $[0, 1] \subseteq \mathbb{R}$ is uncountable.

Proof 2.1.12

Proof by Reduction to Binary Sequences that $[0, 1]$ is Uncountable.

- (1) Since $[0, 1] \supseteq [0, 1)$, it suffices to show $[0, 1)$ is uncountable. Send each $x \in [0, 1)$ to the binary expansion $(b_n) \in 2^{\mathbb{N}}$ that does *not* end in all 1s; on $[0, 1)$ this representative exists and is unique, so the map is injective, with image exactly the sequences not ending in all 1s.
- (2) If $[0, 1)$ were countable, that image would be countable, so $2^{\mathbb{N}}$ —the image together with the countably many all-1-tail sequences—would be countable too (R 2.12), contradicting 2.1.10.
- (3) Hence $[0, 1)$, and therefore $[0, 1]$, is uncountable. ■

[Direct] uses 2.1.10; barring an all-1 tail makes binary expansions unique ($0.0111\dots_2 = 0.1000\dots_2$), and the barred eventually-all-1 sequences are countable.

Corollary 2.1.13

$\mathbb{R}$ is uncountable

Were $\mathbb{R}$ countable, its infinite subset $[0, 1]$ would be countable too (2.1.9); but it is not (2.1.12).

So $\mathbb{R}$ is uncountable.

Remark 2.1.14

$\mathbb{Q}$ is thin in $\mathbb{R}$

Thus $\mathbb{Q}$ is countable yet dense (1.7.2), while $\mathbb{R}$ is uncountable: the rationals are a vanishingly small part of the line. Made precise by measure, $\mathbb{Q}$ —indeed any countable set—has Lebesgue length zero, so a real drawn from any distribution with a density is irrational with probability 1.

2.2 Metric spaces

The Euclidean distance on $\mathbb{R}^n$, $d(x, y) = \|x - y\| = (\sum_{i=1}^n |x_i - y_i|^2)^{1/2}$, built from the inner product $\langle x, y \rangle = \sum_i x_i y_i$, is the model. A metric space keeps only the features of d that analysis actually uses.

Definition 2.2.1

Metric space

A *metric space* (R 2.15) is a set X with a *metric* (or distance) $d : X \times X \rightarrow \mathbb{R}^{\geq 0}$ satisfying, for all $x, y, z \in X$:

1. $d(x, y) \geq 0$, with $d(x, y) = 0 \iff x = y$;
2. $d(x, y) = d(y, x)$ (symmetry);
3. $d(x, z) \leq d(x, y) + d(y, z)$ (the triangle inequality).

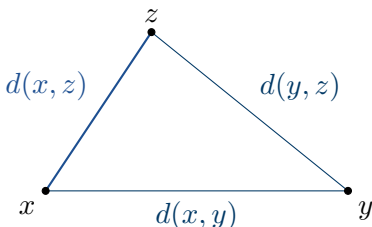


Figure 2.2.2. THE TRIANGLE INEQUALITY. The direct distance $d(x, z)$ never exceeds the detour $d(x, y) + d(y, z)$ through y —axiom (3) of 2.2.1.

Example 2.2.3

The standard metrics

$(\mathbb{R}, |\cdot|)$ with $d(x, y) = |x - y|$ is a metric space, as are $(\mathbb{Z}, |\cdot|)$, $(\mathbb{N}, |\cdot|)$, and $(\mathbb{Q}, |\cdot|)$. More generally any subset Y of a metric space (X, d) is itself a metric space under the restricted distance $d|_{Y \times Y}$. The Euclidean space $(\mathbb{R}^n, d_2)$, $d_2(x, y) = (\sum_{i=1}^n |x_i - y_i|^2)^{1/2}$, is the prototype.

Example 2.2.4

The p -metrics and the discrete metric

On $\mathbb{R}^n$, for $p \geq 1$,¹

$$d_p(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}, \quad d_\infty(x, y) = \max_i |x_i - y_i|.$$

On any set X , the *discrete metric*— $d(x, y) = 0$ if $x = y$ and 1 otherwise—is a metric. The triangle inequality for d_p is Minkowski's inequality, taken here as a standard fact.

Example 2.2.5

Metrics on a function space

Let $C = C[0, 1]$ be the continuous functions $f : [0, 1] \rightarrow \mathbb{R}$. Both

$$d_\infty(f, g) = \sup_{[0, 1]} |f - g| \quad \text{and} \quad d_1(f, g) = \int_0^1 |f(x) - g(x)| dx$$

are metrics on C . Here d_∞ is finite because a continuous function on $[0, 1]$ is bounded, and $d_1(f, g) = 0$ forces $f = g$ since $|f - g|$ is continuous.

2.3 Open sets and open balls

Basic *topology* studies a space through its *open subsets*—concepts that can be phrased using open sets alone. Everything starts with the open ball.

¹The board writes $p > 1$; in fact d_p is a metric for every $p \geq 1$, with d_1 and d_∞ the boundary cases.

Definition 2.3.1

Open ball

In a metric space (X, d) , the *open ball* of radius $r > 0$ about $p \in X$ is

$$B_r(p) = \{x \in X : d(x, p) < r\}.$$

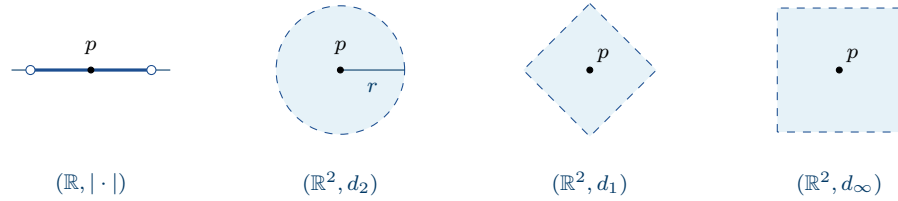


Figure 2.3.2. AN OPEN BALL TAKES THE SHAPE OF ITS METRIC. Same centre, same radius: an interval in $(\mathbb{R}, |\cdot|)$, a round disk under d_2 , a diamond under d_1 , a square under d_∞ —each the set $\{x : d(x, p) < r\}$ of 2.3.1.

Definition 2.3.3

Interior point; open set

Let (X, d) be a metric space and $E \subseteq X$. A point $p \in E$ is an *interior point* of E if $B_r(p) \subseteq E$ for some $r > 0$, and E is *open in X* if every point of E is interior (2.18). Conversely, p fails to be interior exactly when $B_r(p) \not\subseteq E$ for *every* $r > 0$; so E is *not* open as soon as one of its points is not interior.

Remark 2.3.4*Openness is relative*

Openness is meaningful only *relative to the ambient space*: one says “ E is open in X ,” never merely “ E is open.” The same set can be open in one metric space and not in another.

Proposition 2.3.5

Every open ball is open

In any metric space (X, d) , the ball $B_r(p)$ is open in X , for every $p \in X$ and $r > 0$.

Proof 2.3.5

Proof by the Triangle Inequality that Every Open Ball is Open.

- (1) Fix $q \in B_r(p)$; then $d(p, q) < r$, so $r_q := r - d(p, q) > 0$.
- (2) Take any $x \in B_{r_q}(q)$. By the triangle inequality,

$$d(x, p) \leq d(x, q) + d(q, p) < r_q + d(q, p) = r.$$

- (3) Hence $x \in B_r(p)$, so $B_{r_q}(q) \subseteq B_r(p)$. Every $q \in B_r(p)$ is therefore interior, and

$B_r(p)$ is open.

■

[Direct] the template for openness proofs; uses 2.2.1; cf. R 2.19.

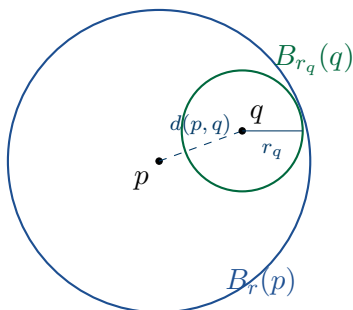


Figure 2.3.6. WHY A BALL IS OPEN. For $q \in B_r(p)$, the radius $r_q = r - d(p, q)$ keeps $B_{r_q}(q)$ inside $B_r(p)$ —the triangle inequality at work (proof of 2.3.5).

Remark 2.3.7

A recipe for proving openness

To show E is open in X : (1) take an arbitrary $p \in E$; (2) produce a radius $r_p > 0$ (guided by the geometry); (3) verify $B_{r_p}(p) \subseteq E$. The proof of 2.3.5 is the model.

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